Mathematics in LaTeX

Compiled mathematical examples

1. Inline and Display Mathematics

$$Inline: a + b = c$$

$$a + b = c$$

2. Superscripts and Subscripts

$$x^2 \; a_1 \; x_1^2$$

$$x^{10} \; a_{12}$$

3. Fractions and Roots

$$a / b$$

$$sqrt(x) \; cube \; root \; of \; x$$

4. Greek Letters

$$alpha, beta, gamma, delta$$

$$pi, theta, lambda, omega$$

5. Mathematical Operators

$$sum \; from \; i = 1 \; to \; n \; of \; i$$

$$product \; from \; i = 1 \; to \; n \; of \; i$$

$$limit \; as \; x \; approaches \; 0 \; of \; f(x)$$

6. Brackets and Grouping

$$(a + b) \; [x + y]$$

$$(a / b)$$

$$[(x + 1) / (x - 1)]$$

7. Equations and Alignment

$$x + 2 = 5$$

$$x = 5 - 2$$

$$x = 3$$

8. Matrices

[1 2]

[3 4]